

Jaehee Hur

AX Data Scientist for Regulated Industries

Seoul, South Korea · [Portfolio](#)

SUMMARY

Senior data scientist with 8+ years building production AI systems end-to-end, from data foundation through model deployment, inside the constrained-data environments of regulated pharma and life-sciences.

Start by identifying the right data for the problem, then design the pipelines that feed it into analysis, with governance and data quality built in from the start. From there, build the automation tools, monitoring dashboards, and ML models that ship to production.

Focused on turning frontline business pain-points into deployed systems that scale analytics adoption across the organization.

KEY CAPABILITIES

- **End-to-end AX delivery:** Led AX projects from business-problem scoping and internal stakeholders pain-points analysis through model development, production deployment, and post-launch maintenance across advanced materials and life-sciences functions at LG Chem.
- **Multi-agent & LLM orchestration:** Built multi-agent pipelines and LLM-based extraction systems for competitive intelligence, regulatory document monitoring, and literature mining, several adopted enterprise-wide.
- **Production deployment & operations:** Shipped AI models and LLM agents to production across GCP and AWS (Cloud Run via GitLab CI/CD; inference serving on EC2), and built local-LLM + RAG pipelines running entirely on on-premise infrastructure for data-residency-constrained environments.
- **Measurable business impact:** Cut multi-week research and review processes to minutes/hours (e.g., 2-week competitor research to under 10 minutes; 23h to 3h per-instrument HPLC review) through deployed AI systems.
- **Cross-functional enablement:** Mentored business users on in-house analytics platforms and led internal AX best-practice sessions, scaling AI adoption beyond the AX team.

EDUCATION

Sookmyung Women's University

M.S., IT Engineering (Data Science)

Seoul, South Korea
Mar. 2015 – Feb. 2017

GPA: 4.43 / 4.5

- Post-M.S. Research Associate, Data Analytics Lab (Mar.–Dec. 2017): ML-based feature-importance methods; patent filings; domestic & international conference papers.

Dongguk University

B.A., International Trade & Advertising/Public Relations (Double Major)

Seoul, South Korea
Mar. 2009 – Feb. 2015

GPA: 3.93 / 4.5

PROFESSIONAL EXPERIENCE

LG Chem — Life Sciences Division, AX Technology Development Team

Senior AX Data Scientist

Seoul, South Korea
Jun. 2022 – Present

- **Lead the end-to-end AX project lifecycle** across advanced material and life-sciences functions: business-problem scoping, internal-customer discovery, data pipeline design, model development, and production deployment.

- **Build and operate analysis pipelines across GCP and Dataiku**, querying BigQuery with SQL to pull data into GCP Workbench and connecting Snowflake datasets into Dataiku, giving analysts ready-to-use environments for downstream analysis and modeling.
- **Deploy AI models and LLM agents to production across GCP and AWS**, shipping 2 in-house agents enterprise-wide to Cloud Run via a GitLab CI/CD pipeline and deploying 4 inference models on an internal ML platform built on AWS EC2.

Selected Projects

- **Company R&D Portfolio Intelligence Agent (2026)**: Built a multi-agent orchestration pipeline that auto-collects and consolidates target companies' financials, news, and patent data into a unified report, cutting per-company research time from 2+ weeks to under 10 minutes. Originally built for R&D portfolio analysis, the tool is now used enterprise-wide across business strategy and marketing functions, generating 800+ reports in its first month, sustaining 20-30 reports per day.
- **Target Compound Literature Screening Platform**: Built an LLM-based extraction platform that pulls experimental conditions, material properties, and thermal stability indicators from target-compound papers and patents into a structured, visualized database. Reduced literature research time for researchers, aimed at supporting the company's goal of shortening material development cycles from 36 months to within 12 months.
- **HPLC Audit-Trail Monitoring Platform (2025)**: Demonstrated, through a PoC across 21 test methods (50,000+ annual HPLC records), a shift from sampling-based review to 100% audit-trail coverage with the potential to cut per-instrument review time from 23h to 3h. Built an anomaly-detection platform with visualization dashboards to flag irregular access patterns, off-hours activity, and abnormally long processing times. Also developed an LLM + RAG chatbot that retrieves answers from updated database of regulatory guidance and internal SOPs.
- **CP Expense Post-Audit Monitoring System (2024-2025)**: Designed a rule-based CP-compliance monitoring system, leading its technical integration into the enterprise OCE system in collaboration with the SI vendor. Expanded monthly MR activity review from under 10% (sampling-based) to 100% coverage across 3,500+ monthly activities, proactively supporting compliance under Korea's Pharmaceutical Affairs Act and Medical Devices Act economic-benefit reporting requirements.
- **Pharmacovigilance Literature Monitoring Platform (2024)**: Built an automated pharmacovigilance adverse-event detection platform by supplementing an open-source LLM-based NER pipeline with sciSpaCy, BioBERT, MedDRA, and rule-based keyword matching to improve extraction of company product names and capture a broader range of entity types. Reduced literature research time for safety officers; shipped as a Streamlit dashboard with Excel export.

GC Holdings — Data Governance Team

Data Scientist

Seoul, South Korea
Dec. 2020 – Jun. 2022

- Identified AX opportunities and synergies across 5+ subsidiaries (GC Healthcare, Ubicare, GC Labs, GC Wellbeing) for holding-company-level data integration, conducting DX discovery interviews and pain-points analyses to derive prioritized task roadmaps.
- Tracked data policy/regulatory shifts (Data 3 Act) and analyzed the digital-healthcare market landscape, including competitor technology portfolios, to identify and pursue new-business directions for data monetization at a pharma/healthcare holding company.
- Enabled compliant healthcare data utilization across the group by serving as pseudonymization officer, contributing to data-governance guideline development and building public-data collection pipelines.

Seoul Asan Hospital — Asan Institute for Life Sciences, Cardiology

AI Research Scientist

Seoul, South Korea
Jan. 2019 – Nov. 2020

- Enabled cardiovascular outcome research and built the foundational dataset for AI-based clinical decision support by constructing a retrospective EMR cohort of 570,000 cardiac patients (2000-2016) from Asan Medical Center's clinical data systems, resolving code fragmentation across patient demographics, diagnosis, surgical procedures, and medication records—extracting and de-identifying both structured fields and unstructured clinical notes via NLP.

- Reduced clinician diagnostic review time for tympanic-membrane conditions by developing a multi-class deep-learning classifier (VGG16/ResNet/DenseNet/EfficientNet ensemble; 95.8% accuracy) with a clinician-facing validation UI and Grad-CAM activation maps.

SKILLS

Deployment & LLM: LangChain / LangGraph, RAG, Local/on-prem LLM(Ollama), Multi-agent orchestration.

Data platforms: GCP (BigQuery, Vertex AI), AWS (EC2, Bedrocks), Dataiku, Snowflake

Engineering & front-end: Python, MySQL, PyTorch, Keras, Scikit-learn; Flask, Streamlit, Plotly Dash; REST API integration; PyInstaller packaging.

Languages: Korean (Native) · English (Intermediate) · French (DELF B1).

Certifications: AICE Associate · ADsP (Data Analysis Semi-Professional).

PUBLICATIONS

- Choi Y., Chae J., Park K., **Hur J.**, Kweon J., Ahn J. (2022). Automated multi-class classification for prediction of tympanic membrane changes with deep learning models. *PLOS ONE*.
- Kwon O., Na W., **Hur J.**, et al. (2021). Cardiovascular Event Rates in Statin-Treated Korean Patients with Cardiovascular Disease. *Cardiovascular Drugs and Therapy*.
- Yang S., Kweon J., Roh J-H., ... **Hur J.**, et al. (2019). Deep learning segmentation of major vessels in X-ray coronary angiography. *Scientific Reports*.
- **Hur J.**, Ihm S., Park Y. (2017). A Variable Impacts Measurement in Random Forest for Mobile Cloud Computing. *Wireless Communications and Mobile Computing*.